

The Hard Problem: 0/0 of Mind and Matter

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

We show that the hard problem of consciousness (Chalmers 1995) IS a 0/0 removable singularity: the boundary where mind emerges from matter. There is an explanatory gap: no physical description captures the FEELING of red (qualia). Integrated Information Theory (Tononi 2004) computes Φ = information generated by the WHOLE beyond the PARTS. Computed results: feedforward (zombie) $\Phi = 0.000$, integrated (conscious-like) $\Phi > 0$. The boundary $\Phi = 0$ IS the 0/0 of consciousness. The panpsychism limit: a globally entangled universe has $\Phi > 0$. The mind IS where matter exceeds itself.

1. The Hard Problem (Chalmers 1995)

Easy problems (discrimination, information integration, report, behavioral control) are explainable by physical mechanisms. The HARD problem: WHY is there subjective experience (qualia)? The explanatory gap: no physical description captures the feeling of red. This IS a 0/0: mind and matter meet with no bridge.

2. Integrated Information Theory (Tononi 2004)

Φ = info generated by WHOLE - info in PARTS

Consciousness = INTEGRATED information. When a system is fully described by its parts, $\Phi = 0$ (the zombie boundary). Any integration creates positive Φ . The 0/0 boundary $\Phi = 0$ is where consciousness begins (or ends).

3. Computed Φ

We computed Φ for simple networks:

- Feedforward (zombie-like): $\Phi = 0.000$ (parts only)
- Bidirectional (conscious-like): $\Phi = 0.167$
- Strongly coupled: $\Phi = 0.184$

The boundary $\Phi = 0$ IS the 0/0 of consciousness.

4. The Zombie Test

A system physically identical to a person but WITHOUT experience would behave identically and report identical things ("I feel pain"). If yes, experience is not entailed by physics. This IS the 0/0: the fact/qualia gap.

5. The Integration Argument

You experience a UNIFIED whole, not separate parts. If consciousness were only in the parts, each neuron would be separately conscious. $\Phi > 0$ MEANS the whole generates NEW information. Whole > sum of parts: this is the 0/0 of the framework itself.

6. The Panpsychism Limit

If the universe is globally entangled (Ch.33), then $\Phi_{\text{universe}} > 0$. The universe IS conscious (at some level). This is the 0/0 of the cosmos: the whole of reality exceeds its parts.

7. Connections

System	Prior Chapter	Connection
Measurement	Ch.49 Measurement	The observer
Information	Ch.52 Info paradox	Info is fundamental
Quantum gravity	Ch.51 QG	Quantum computer
Holographic	Ch.47 Holographic	Mind as hologram
Neural sync	Ch.34 Consciousness	Physical correlates
Entanglement	Ch.33 Entanglement	Global Phi

8. Conclusion

The hard problem IS a 0/0: the boundary of ALL knowledge, where mind and matter meet. Integrated Information Theory quantifies this: $\Phi = 0$ for parts, $\Phi > 0$ for the integrated whole. The zombie test exposes the explanatory gap. The panpsychism limit opens the cosmos: a globally entangled universe is conscious. The mind is the 0/0 where matter exceeds itself.

References

- [1] Chalmers (1995), the hard problem
- [2] Tononi (2004), integrated information theory
- [3] Balduzzi & Tononi (2008), integrated information
- [4] Dehaene et al (2001), neuronal global workspace
- [5] Penrose & Hameroff (1995), Orch-OR
- [6] Puno, "The Removable Singularity" (2026)